# From de Bruijn Graphs to Rectangle Graphs for Genome Assembly

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**Abstract.** Jigsaw puzzles were originally constructed by painting a picture on a rectangular piece of wood and further cutting it into smaller pieces with a jigsaw. The *Jigsaw Puzzle Problem* is to find an arrangement of these pieces that fills up the rectangle in such a way that neighboring pieces have "matching" boundaries with respect to color and texture. While the general Jigsaw Puzzle Problem is NP-complete [6], we discuss its simpler version (called *Rectangle Puzzle Problem*) and study the *rectangle graphs*, recently introduced by Bankevich et al., 2012 [3], for assembling such puzzles. We establish the connection between Rectangle Puzzle Problem and the problem of assembling genomes from read-pairs, and further extend the analysis in [3] to real challenges encountered in applications of rectangle graphs in genome assembly. We demonstrate that addressing these challenges results in an assembler SPAdes+ that improves on existing assembly algorithms in the case of bacterial genomes (including particularly difficult case of genome assemblies from single cells).

SPAdes+ is freely available from http://bioinf.spbau.ru/spades.

## 1 Introduction

The recent proliferation of next generation sequencing technologies has enabled new experimental opportunities and, at the same time, raised formidable computational challenges. When the length of a repeat in the genome exceeds the read length, it becomes difficult to "span" the flanking regions of this repeat in the assembly. To alleviate this problem, sequencing technologies were extended to produce *read-pairs*, pairs of reads separated by an estimated *insert length*. Because insert length is longer than the read length, read-pairs span longer repeats and could potentially result in better assemblies. However, while assembling *single* reads can be elegantly modeled by *de Bruijn graphs* [7], equally elegant models for assembling *read-pairs* remain unknown [6].

Pevzner and Tang, 2001 [11] addressed this challenge by constructing the de Bruijn graph and further checking if a path between two reads within a read-pair satisfies the constraint imposed by the insert length. If only one such path exists,

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the read-pair is transformed into a virtual long read where the gap between reads is filled in with the nucleotide sequence representing the found path.

While this and similar methods [9, 13] had a large impact on genome assembly, they fail in repeat-rich regions, where there are multiple paths between the reads within read-pairs. Recently, Medvedev et al., 2011 [10] introduced *paired de Bruijn graphs* that directly incorporate read-pairs into the graph structure and bypass the problem of multiple paths in previous approaches. However, the paired de Bruijn graph concept was introduced as a theoretical framework and is mainly aimed at an unrealistic case when the distance between reads within read-pairs is exactly $d$ for all read-pairs. To address this bottleneck, Bankevich et al., 2012 [3] introduced the *rectangle graph* by generalizing the problem of a string reconstruction from its paired substrings to a variation of a jigsaw puzzle problem. While fragment assembly is usually modeled as a 1-dimensional *overlapping* puzzle (pieces correspond to individual reads), Bankevich et al. [3] modeled assembly as a 2-dimensional *non-overlapping* puzzle (pieces correspond to pairs of paths in the de Bruijn graph). However, while Bankevich et al. [3] sketched the rectangle graph idea, the various questions arising in applications of rectangle graphs to fragment assembly remained unaddressed.

The jigsaw puzzles were originally constructed by painting a picture on a rectangular piece of wood and further cutting it into smaller pieces with a jigsaw. The *Jigsaw Puzzle Problem* is to find an arrangement of these pieces that fills up the rectangle in such a way that neighboring pieces have "matching" boundaries with respect to color and texture. This paper extends the previous algorithmic studies of the Jigsaw Puzzle Problem (that were motivated by the restoration of archaeological artifacts [8]) to the problem of genome assembly from read-pairs. In section 2, we describe a class of simple jigsaw puzzles (called *rectangle puzzles*) and define the rectangle graph to assemble such puzzles. In section 3, we establish the relation between the rectangle puzzle and genome assembly from read-pairs and address the algorithmic challenges of "missing rectangles" (that often arise in genome assembly) in the rectangle puzzle problem. In Section 4, we apply the rectangle graph to bacterial genome assembly for both standard (multicell) and more difficult single cell datasets.

## 2 Rectangle Puzzles

Consider $n+1$ points $x_0 = 0 < x_1 < \ldots < x_n$ on $x$-axis and $m+1$ points $y_0 = 0 < y_1 < \ldots < y_m$ on $y$-axis. Points $(x_i, y_j)$ form a 2-dimensional grid consisting of $n \cdot m$ rectangles filling up the grid with corners $(0,0), (0,y_m), (x_n,0)$ and $(x_n, y_m)$. By analogy with the jigsaw puzzle assembly, we assume that the grid is "painted" and the goal is to assemble small rectangles into the painted grid in such a way that rectangles fully fill up the grid and that the colors at the sides of neighboring rectangles match (*valid assembly*). To simplify the matters, we will assume that the "orientation" of each rectangle is known.

Assembling the Ha Long Bay puzzle in Fig. 1a is trivial (for every rectangle, there exists an unambiguous choice of neighboring rectangles). Fig. 1b shows 9 rectangles from this puzzle with 12 blue dotted lines connecting the *unique*